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Artificial Intelligence Applications for Predicting Discharge Pressure and Flowrate of Electrical Submersible Pumps for Performance Optimization and Failure Prevention

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Abstract

The natural production of most oil wells inevitably declines over time, therefore, artificial lift methods are essential to maintain oil production. Among these methods, electrical submersible pumps (ESPs) are reliable and demonstrate a high level of productivity. However, ESPs are subjected to degradation or failure during their run life, leading to production losses. This study utilizes artificial intelligence (AI) to optimize the performance of ESPs using real field data from an Arab oil field.

The study focuses on dual-parameter prediction for Pump Discharge Pressure (P_d) and Flow Rate (Q). Their impact on the run life of the ESP system was evident through trend analysis when compared to other features. The dataset for this study—comprising over 12,000 datapoints from 44 oil wells—was collected using real-time monitoring and includes 18 parameters such as surface data, reservoir and fluid properties, and pump specifications. During pre-processing, the Pearson Correlation Coefficient (PCC) indicated data non-linearity. Consequently, Support Vector Regression (SVR) and Extreme Gradient Boosting (XGBoost) were employed due to their effectiveness in managing such complexities. Additionally, a Random Forest model was utilized to predict ESP system failures.

The performance of each model was evaluated using key metrics, including the coefficient of determination (R^2), Root Mean Squared Error (RMSE), and Average Absolute Percentage Error (AAPE). For the SVR model, the testing data yielded R^2 values of 0.955 for P_d and 0.917 for Q , with corresponding RMSE values of 76.2 and 157.2, and AAPE values of 1.01% for P_d and 5.92% for Q . In comparison, XGBoost demonstrated superior performance, achieving R^2 values of 0.968 for both P_d and Q , RMSE values of 64.16 and 98.16, and AAPE values of 0.84% for P_d and 3.48% for Q . The classification model successfully predicted failures during testing and was evaluated using precision, recall, and confusion matrix, achieving an overall accuracy of 96%. A comparative analysis of regression models was conducted, and XGBoost was identified as the most accurate model, delivering the best results. Ultimately, the developed models collectively aim to optimize company resources by predicting flowrates during recording challenges and by allocating personnel efforts to ESPs with a high risk of failure.

In summary, traditional monitoring methods for ESPs (such as vibration analysis) are reactive and often detect problems only after failures occur. These limitations highlight the need for advanced monitoring solutions. Leveraging AI to optimize ESP performance is critical for ensuring sustainable and cost-effective production. The research findings can be applied across various ESPs, enabling proactive maintenance strategies that minimize downtime, reduce energy consumption, and lower operational costs. This study significantly contributes to the existing body of literature by providing a novel approach to optimizing ESP performance.

Introduction

The natural production of most oil wells declines over time, necessitating the use of artificial lift systems to sustain or enhance production. Several artificial lift methods are commonly used in the oil industry, including electrical submersible pumps (ESP), gas lift, progressive cavity pumps (PCP), and sucker rod pumps. Among these, ESP and gas lift are the most widely applied in high-rate wells (Clegg 1988). While suitable for different wellbore and fluid conditions, ESP systems are often more efficient and reliable for lifting moderate to large fluid volumes in low-pressure, high-water cut reservoirs (Bearden 2007; Islam 2005).

ESP systems are prone to failure during their run life due to electrical or mechanical issues, which can result in production losses, increased workover frequency, and higher energy consumption. Routine monitoring and maintenance are therefore essential to maintain optimal performance (Bremner et al. 2006). Traditional monitoring methods—such as current tracking and vibration analysis—are reactive, prone to false alarms, and often identify faults only after failure has occurred. These limitations underscore the need for more advanced, predictive solutions.

Artificial intelligence (AI) has proven effective in numerous oil and gas applications. This study applies machine learning techniques to enhance ESP performance and enable predictive maintenance by training models on real field data. Recent innovations in ESP design have improved system tolerance to high temperatures and increased gas handling capacity, up to 75% gas volume fraction (Bremner et al. 2006; Alsinan et al. 2024). However, gas interference in flowlines remains a significant operational challenge. Additional concerns, including overheating, corrosion, and material degradation, also demand attention to ensure production reliability.

Employing AI to forecast ESP failure and assess key performance factors offers a robust, data-driven approach to extend system life and minimize operational risk. This study develops machine learning models capable of predicting discharge pressure and flow rate, two primary indicators of ESP performance. These predictions support proactive decision-making for maintenance and pump operation, ultimately reducing downtime and improving both cost efficiency and operational safety.

The general objective is to utilize machine learning techniques for Electrical Submersible Pump (ESP) systems to enhance operational reliability by forecasting failures, detecting performance degradation, reducing downtime, and optimizing cost-efficiency. The detailed objectives of this research are to predict key ESP performance indicators (discharge pressure and flow rate) to support real-time monitoring of pump behavior; to provide a data-driven solution that minimizes reliance on physical flowmeters and production testing, particularly in oilfields with high equipment and maintenance costs; to implement a failure prediction framework for the early identification of ESP failures by analyzing deviations between predicted and observed pressure and flow rate values; and to evaluate and compare the predictive performance, accuracy, and reliability of Support Vector Regression (SVR), Extreme Gradient Boosting (XGBoost), and Random Forest models in forecasting ESP discharge pressure and flow rate using real field data under varying operating conditions.

Methodology

The performance of ESPs was evaluated under a range of reservoir conditions, spanning from low to high bottom hole pressures, reflecting varying levels of reservoir productivity and energy, with both gas-oil ratios (GOR) and water cut (WC) exhibiting low-to-moderate values. The analysis also revealed that the extracted oil is heavy, with an average API gravity of 19.2 °, and is produced from a carbonate reservoir. These facts highlight the importance of robust optimization strategies to enhance ESP performance in similar environments. Figure 1 presents a flowchart that outlines the step-by-step methodology followed throughout the project.



Figure 1—Flowchart illustrating the step-by-step methodology followed throughout the project.

Data Collection and Description

The data was collected from an Arab Oil Field with an extending carbonate formation, utilizing a real-time ESP sensor to simultaneously record various downhole parameters. Expert engineers on-site, who were familiar with the real-time sensor and its application, carefully monitored the data collection process to ensure accurate recording of all necessary parameters. The acquired dataset included 18 parameters recorded across 62 wells, encompassing both oil and water wells' data. Since the study aims to optimize ESP performance in oil production, water wells were excluded from the dataset, along with wells that had insufficient data—to maintain the reliability and accuracy of the trained ML models. Consequently, nine water wells and seven oil wells (those with insufficient data) were excluded, resulting in 44 oil wells and

approximately 24,000 raw data. The dataset represents information on surface data, reservoir properties, fluid properties, and pump specifications.

Reservoir Characteristics

The data utilized in this study were collected from a field in the Middle East, characterized by its diverse geological formations. An analysis was conducted to classify and understand each reservoir's characteristics, specifically lithology. The seven studied formations, designated as formations A, B, C, D, E, F, and G, are classified in the literature as carbonate formations, with the majority of the data collected from formation A. This classification is significant, as carbonate formations are generally associated with reduced risks of sand production due to their consolidated structure. As a result, none of the wells analyzed in this study exhibited sand production issues.

Moreover, bottom hole pressure (BHP) data from multiple operating oil wells were collected and analyzed to evaluate reservoir productivity. Descriptive statistical analysis was performed to determine the distribution of BHP values, including the minimum, maximum, mean, median, and standard deviation. The results indicated a wide range of bottom hole pressures, with values spanning from 154 psi to 5,576 psi. The mean BHP was approximately 2,830 psi, while the median was 2,848 psi, suggesting a relatively symmetric distribution which indicates most of the wells have moderate reservoir pressure. A standard deviation of approximately 1,101 psi reflects significant variability among wells, indicating differing reservoir conditions, with some wells exhibiting high reservoir pressure and others showing much lower pressure due to depletion or reduced productivity.

Since the dataset from 44 wells includes information about the density of the produced oil from each well, it was possible to classify the type of oil being extracted. API gravity is crucial in determining the performance and efficiency of Electrical Submersible Pumps (ESPs) in oil production. The type of oil—whether light, medium, heavy, or extra-heavy—affects factors such as viscosity, flowrate, and the load on the pump. To better understand these impacts, an analysis was conducted to highlight the variations in API gravity across the wells, determine the higher and lower values, and calculate the average value.

As illustrated in [Figure 2](#), the API gravity of oil from 44 wells ranges from 14.4° to 30.9° API, with an average value of 19.2° API. According to [Peters et al. \(2012\)](#), oil is classified as light if its API gravity exceeds 31.1° API and heavy if it is below 22.3° API. Based on this classification, the oil in question is categorized as heavy. This observation highlights the importance of monitoring ESP performance, as the high viscosity associated with heavy oil can result in significant performance degradation. Supporting this, a numerical study by [Ofuchi et al. \(2017\)](#) simulating flow in a mixed-flow ESP operating with varying fluid viscosities and flowrates demonstrated that viscosity has a substantial impact on head and flowrate.

Monitoring water cut (WC) is crucial in wells operating with Electrical Submersible Pumps (ESPs), as the presence and properties of water directly impact pump performance and longevity. High water cut can reduce pump efficiency by altering fluid viscosity and increasing energy consumption, while also contributing to operational instabilities through emulsion formation. Moreover, the presence of gas within the produced water can exacerbate these issues by inducing cavitation, which may lead to severe mechanical damage. Therefore, early identification and control of water cut are essential for optimizing ESP operation, minimizing maintenance requirements, and ensuring stable and efficient oil production. In this study, descriptive statistical analysis revealed a wide range of water cut values, spanning from 0% to 100%. The mean water cut was approximately 33.08%, while the median value was 23%, indicating that most wells produced relatively low amounts of water, with a minority of high water cut wells skewing the average upward.

Also, gas-oil ratio (GOR) is a critical indicator of reservoir fluid characteristics, and its impact on ESP performance must be carefully monitored. Statistical analysis of the available GOR data revealed a mean of 322.23 scf/STB, a median of 341, and a mode of 0, with values ranging from 0 to 1,239 scf/STB. The dataset exhibits moderate variability with a standard deviation of 213.07 and slight positive skewness,

indicating that most values are clustered below 500, with a few higher outliers. According to standard industry classifications, the majority of wells fall within the moderate GOR category, while some exhibit high GOR values, suggesting the production of volatile oil. Dissolved gas at these levels can affect ESP performance by reducing pump efficiency, increasing the risk of gas lock or cavitation, and demanding additional gas handling equipment.

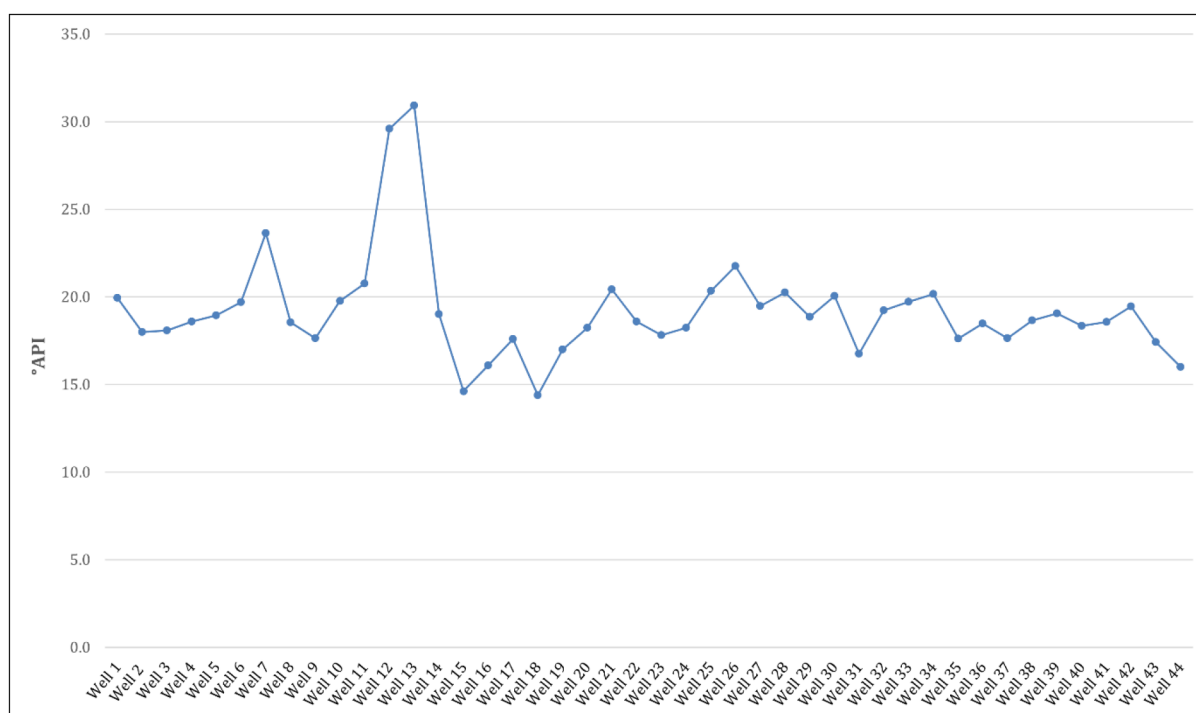


Figure 2—Oil gravity (API) trend across the 44 wells.

Wells Categorization

The dataset collected from 44 oil wells covered approximately 443 months, equivalent to 12,500 operational days. As observed, there was considerable variability in data density across the wells, with certain wells exhibiting substantially more comprehensive records than others. To address the uneven distribution, a systematic categorization was implemented based on the number of operating months associated with each pump. This approach ensured that wells with more data were prioritized for in-depth analysis, enabling a more reliable evaluation of the 18 studied ESP parameters. The wells were systematically classified into three categories based on the length of their recorded data. Category 1, considered poor in terms of data quality, included wells with only 1–10 months of data (~0.5 years of operation). Category 2, representing a good quality range, comprised wells with 11–20 months of data (~1.3 years). Category 3, deemed excellent for analysis, consisted of wells with 21–30 months of data (~2.2 years). This classification established a clear and structured framework that enhanced the integrity and interpretability of the analysis.

Based on the categorization, nine wells were classified into Category 3, accounting for 54% of the total dataset. The significant representation of these wells highlighted their critical role in conducting investigations into anomalies. Furthermore, the substantial amount of data among the nine wells made them well-suited for trend analysis, enabling the identification of the most fluctuating parameters and offering valuable insights into potential issues that could hinder ESP system performance.

Target Parameter Selection

The trend analysis served as the initial selection principle for identifying potential parameters. Four wells from Category 3 were selected due to their extensive data records, which provided valuable insights into

operational trends and parameter behavior over time. The analysis employed Microsoft Excel as the primary tool for creating individual visualizations for each parameter across all wells—involving plotting versus time to examine progressive fluctuations in parameters—resulting in a set of 72 plots.

Each parameter was compared to its recommended operational range, as specified in ESP manuals and handbooks. The comparison confirmed that no parameter was operating outside its specified range. Moreover, after conducting the trend analysis, certain parameters were observed to fluctuate significantly. Anomalies were highlighted for further investigation, with a specific focus on the following parameters: Discharge Pressure (P_d), Flowrate (Q), Motor Temperature (T_m), and Tubing Head Pressure (THP). The four parameters are critical factors that significantly impact Electric Submersible Pump (ESP) performance and longevity.

Previously, discharge pressure was identified in the literature as a critical parameter and therefore selected for further investigation (Mahmoud et al. 2023). On the other hand, flowrate directly affects ESP performance, with higher rates generally leading to better efficiency. However, excessive gas content can cause performance deterioration and surging, especially at lower liquid flowrates (Gamboa and Prado 2012). Motor temperature is crucial for ESP run life and reliability. Elevated temperatures can deteriorate the insulating properties of the electrical system, leading to increased current leakage over time, hence, proper motor cooling is essential (Alrashidi et al. 2020). Tubing head pressure affects the overall system performance by influencing the net hydraulic load and power consumption of the ESP installation (Powers 1988). In conclusion, careful monitoring and management of these parameters are essential for maximizing ESP performance, efficiency, and run life.

Following the trend analysis, a further filtering step was introduced to narrow down the parameters for further evaluation. This criterion focused on prioritizing the most critical parameters that directly impact key aspects of ESP functionality—including pump performance, energy consumption, and operational stability—based on their physical properties, operational impact, and significance in optimizing ESP operations. Ultimately, among the four parameters, Discharge Pressure (P_d) and Flow Rate (Q)—as illustrated in Figure 3 and Figure 4, respectively—were identified as the most significant factors. P_d represents the pump's ability to lift fluids to the surface, while Q indicates the ESP's capacity to achieve production targets and sustain reservoir drawdown. The selected parameters are then modeled using each of the following machine learning algorithms: Support Vector Regression (SVR), Extreme Gradient Boosting (XGBoost), and Random Forest.

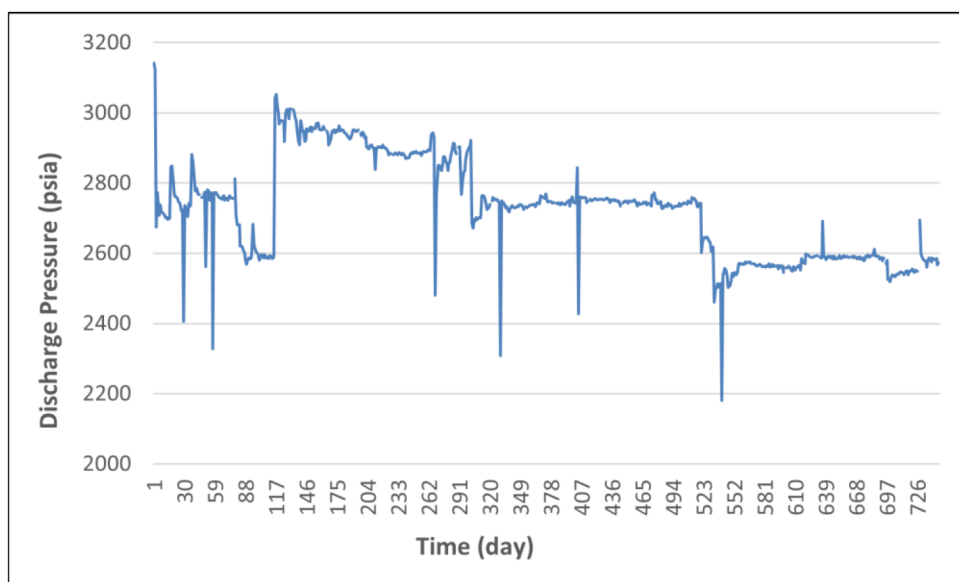


Figure 3—Pump discharge pressure (P_d) trend analysis

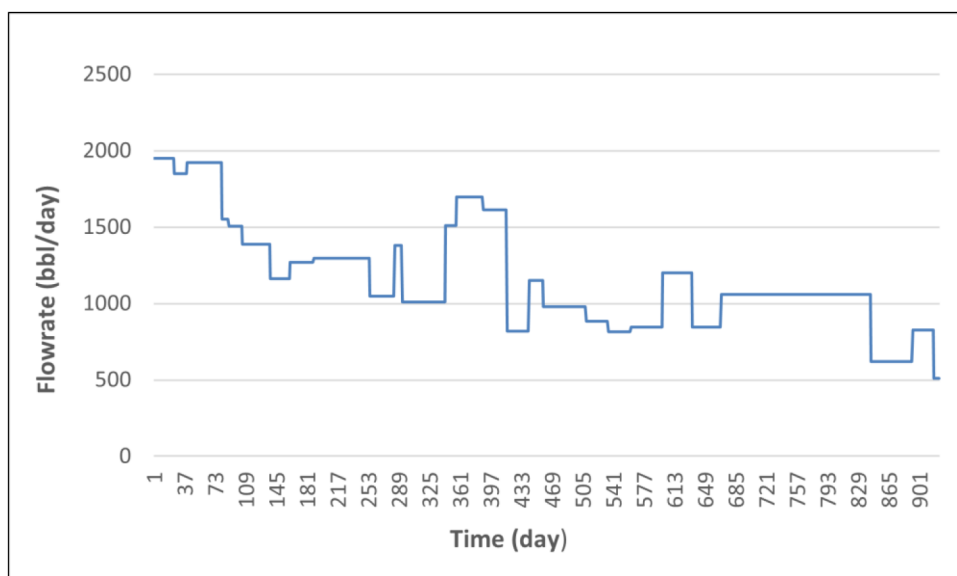


Figure 4—Flow rate (Q) trend analysis

Machine Learning Algorithms

The selection of appropriate machine learning models for this study was guided by the nature of the target variables—discharge pressure (P_d) and flowrate (Q)—and the non-linear, multi-dimensional characteristics of the dataset derived from 44 heavy oil wells. After performing exploratory data analysis and correlation studies, it became evident that conventional linear regression models would not adequately capture the complexity of the relationships between input features and the target parameters.

Therefore, a literature review was conducted to identify algorithms known for their ability to model non-linear behavior while maintaining interpretability and robustness. Support Vector Regressor (SVR) and XGBoost Regressor were selected due to their proven performance in regression tasks involving noisy, high-dimensional data, particularly in engineering applications. Random Forest was also selected for its ensemble learning capabilities and versatility, which made it a suitable choice for classification tasks such as failure detection.

Exploratory Data Analysis (EDA)

Komorowski et al. (2016) discuss that exploratory data analysis (EDA) is an essential step in any research analysis. The main goal of exploratory analysis showed in Table 1 is to investigate the data for its distribution, as well as to identify any outliers and anomalies. The summary below provides insights into the central tendencies, variability, and distribution of all 18 parameters. The statistics include the mean, median, standard deviation, range (minimum to maximum), and the 25th and 75th percentiles.

Table 1—Statistical summary of key ESP operating parameters including inputs and outputs

ESP Parameter	Unit	Mean	Median	Standard Deviation	Minimum	Maximum	25th Percentile	75th Percentile
Discharge pressure (P_d)	psi	3023.1	2972.0	397.1	1190.0	4726.0	2813.0	3215.0
Flowrate (Q)	bbl/day	1305.4	1295.0	557.3	212.0	4323.0	841.0	1795.0
Oil gravity	API	19.6	19.0	3.08	14.2	35.0	17.8	19.9
Casing head pressure (CHP)	psi	180.8	180.0	61.8	0.0	740.0	150.0	210.0
Choke size	1/64"	27.7	26.0	7.1	10.0	52.0	22.0	32.0
Downhole fluid level (DFL)	m	1108.9	1078.1	278.0	0.0	2270.7	945.7	1285.8
ESP amps	A	28.8	28.9	6.2	18.0	328.0	24.0	32.0
ESP voltage	V	412.0	312.0	347.6	225.0	4469.9	274.0	470.0
Flow line pressure (FLP)	psi	179.4	178.0	25.2	0.0	365.0	160.0	198.0
Flow line temperature (FLT)	°C	55.4	56.0	10.5	19.0	81.0	48.0	63.0
Frequency	Hz	40.1	40.0	3.2	35.0	52.0	38.0	42.0
Intake pressure (P_i)	Psi	2040.3	2009.0	317.8	1215.0	3426.0	1828.0	2163.0
Motor temperature (T_m)	°C	98.9	96.5	9.4	74.1	129.1	92.8	103.1
Pump intake temperature (T_i)	°C	90.6	91.0	5.1	71.4	100.2	87.2	94.1
Submergence	m	1172.0	1143.9	224.4	547.5	2341.8	1042.3	1257.5
Tubing head pressure (THP)	Psi	514.9	510.0	130.9	35.0	1100.0	400.0	610.0
Vibration	g	0.42	0.36	0.24	0.0	2.2	0.26	0.51
VSD amperage	A	147.7	129.0	58.9	19.0	406.0	108.0	197.0

The Pearson Correlation Coefficient (PCC) was employed to measure the strength and direction of linear relationships between input parameters and target variables (P_d and Q), and further illustrated with the correlation heatmap in Figure 5. However, the analysis revealed that the overall input-output relationships are nonlinear. As a result, we prioritized machine learning models capable of capturing this complexity.

Specifically, SVR, XGBoost, and Random Forest were selected for their effectiveness in modeling nonlinear relationships.

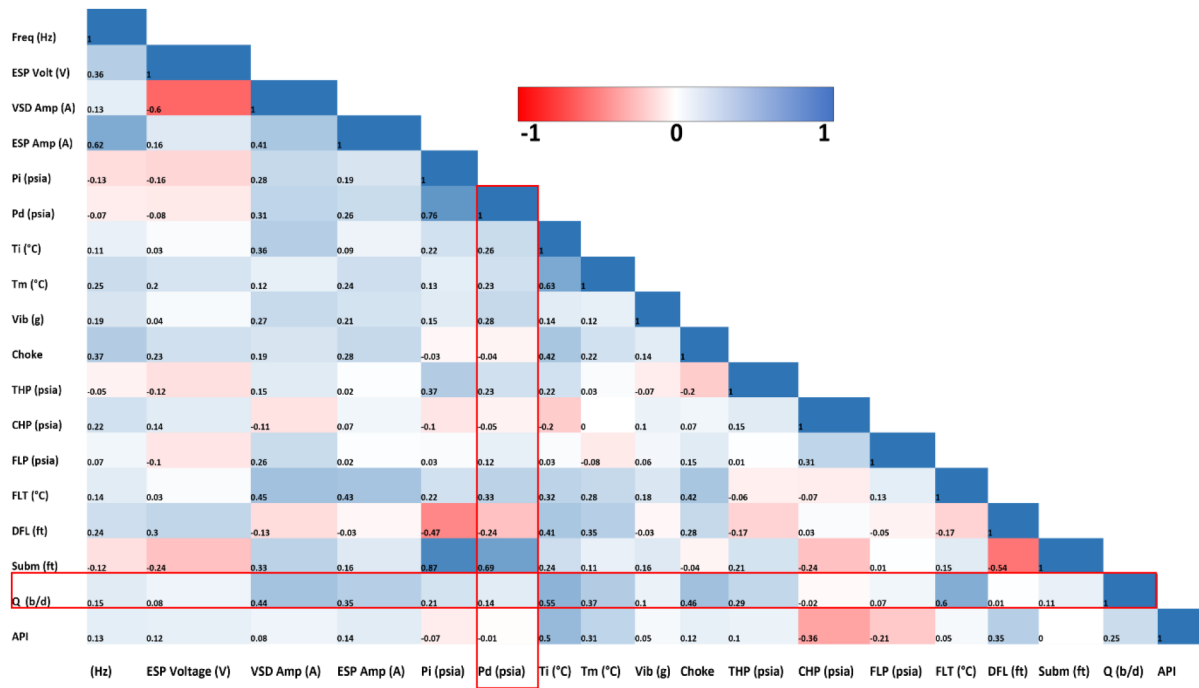


Figure 5—Heatmap of Pearson Correlation Coefficients (PCC) among input features, illustrating the strength and direction of linear relationships between variables

In Table 2, the PCC values quantify the strength of linear relationships between various ESP parameters and the target variables. Parameters with strong positive or negative correlations indicate significant influence, whereas weak correlations suggest minimal impact. The interpretation column provides insights into the role of each parameter in ESP performance.

At this stage, the data distribution was visualized through histogram plots, with Figures 6 through 9 illustrating samples of key ESP parameters. Figure 6 shows a symmetrical distribution of discharge pressure, while Figure 7 reveals a left-skewed pattern in flowrate. Figure 8 displays the distribution of Tubing Head Pressure (THP), highlighting the data spread and indicating a slight right-skewed trend. Similarly, Figure 9 illustrates the distribution of intake temperature (T_i), which also exhibits a right-skewed pattern.

Table 2—Correlation coefficient analysis of ESP parameters in relation to discharge pressure (P_d) and flowrate (Q)

ESP Parameter	PCC Value (P_d)	Linearity (P_d)	PCC Value (Q)	Linearity (Q)	Interpretation
Discharge pressure (P_d)	1.00	Strong (+)	0.14	Weak (+)	The target variable exhibits self-correlation and has a weak association with Q .
Flowrate (Q)	0.14	Weak (+)	1.00	Strong (+)	The target variable exhibits self-correlation and has a weak association with P_d .
Oil gravity (API)	-0.01	Weak (-)	0.25	Weak (+)	It has no measurable effect on P_d but has a slight impact on Q , indicating that lighter fluids with higher API gravity may flow more easily.
Casing head pressure (CHP)	-0.05	Weak (-)	-0.02	Weak (-)	Weak influence on both targets.
Choke size	-0.04	Weak (-)	0.46	Strong (+)	A major driver of Q (a larger choke reduces restriction) with no impact on P_d .
Downhole fluid level (DFL)	-0.24	Weak (-)	0.01	Weak (+)	Low fluid levels reduce P_d but do not restrict Q .
ESP amps	0.26	Weak (+)	0.35	Weak (+)	Similar to VSD amperage, higher amperage supports pump performance.
ESP voltage	-0.08	Weak (-)	0.08	Weak (+)	Weak influence on both targets.
Flow line pressure (FLP)	0.12	Weak (+)	0.07	Weak (+)	Weak influence on both targets.
Flow line temperature (FLT)	0.33	Weak (+)	0.60	Strong (+)	Thermal expansion significantly increases Q by reducing viscosity, with a moderate impact on P_d .
Frequency	-0.07	Weak (-)	0.15	Weak (+)	Higher speed increases Q with weak effect on P_d .
Intake pressure (P_i)	0.76	Strong (+)	0.21	Weak (+)	It dominates P_d due to the direct hydraulic link but has weak impact on Q .
Motor temperature (T_m)	0.23	Weak (+)	0.37	Weak (+)	T_m is more critical for Q than P_d , indicating motor stress is flow-driven.
Pump intake temperature (T_i)	0.26	Weak (+)	0.55	Strong (+)	Similar to FLT, heat reduces fluid resistance, enhancing Q while having minimal effect on P_d .
Submergence	0.69	Strong (+)	0.11	Weak (+)	Higher submergence increases intake pressure (P_i), leading to improved P_d , but weakly effect on Q .
Tubing head pressure (THP)	0.23	Weak (+)	0.29	Weak (+)	Slight hydraulic influence on P_d as it also facilitates fluid flow.
Vibration	0.28	Weak (+)	0.10	Weak (+)	Any increase in vibration indicates mechanical wear that may destabilize P_d but has minimal impact on Q .
VSD amperage	0.31	Weak (+)	0.44	Strong (+)	Higher current draws increase pump work and speed, leading to higher P_d and Q , respectively.

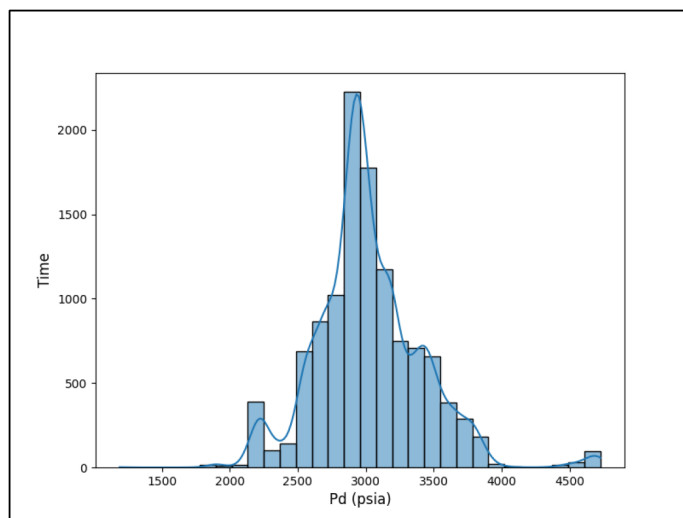


Figure 6—Histogram plot illustrating the distribution of discharge pressure (P_d).

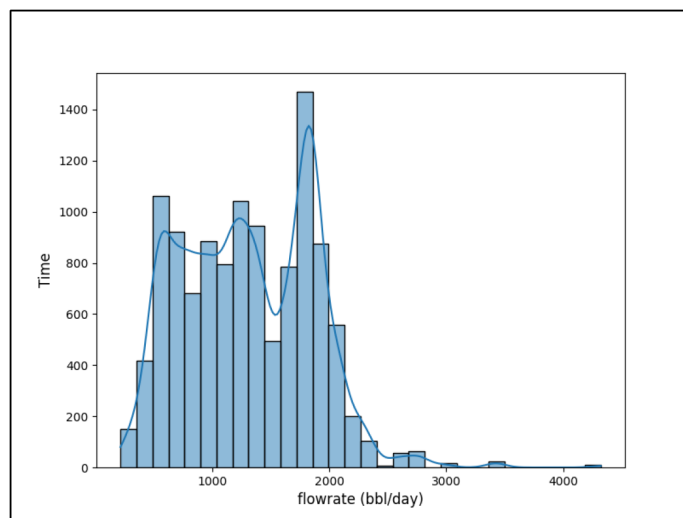


Figure 7—Histogram plot illustrating the distribution of flowrate (Q).

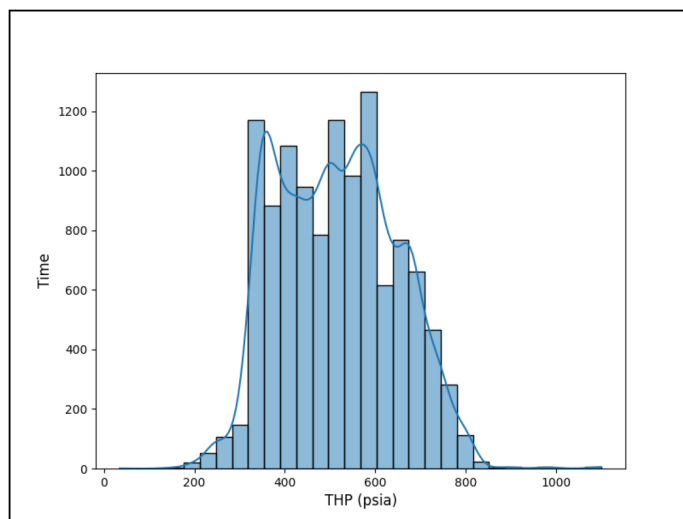


Figure 8—Histogram plot illustrating the distribution of tubing head pressure (THP).

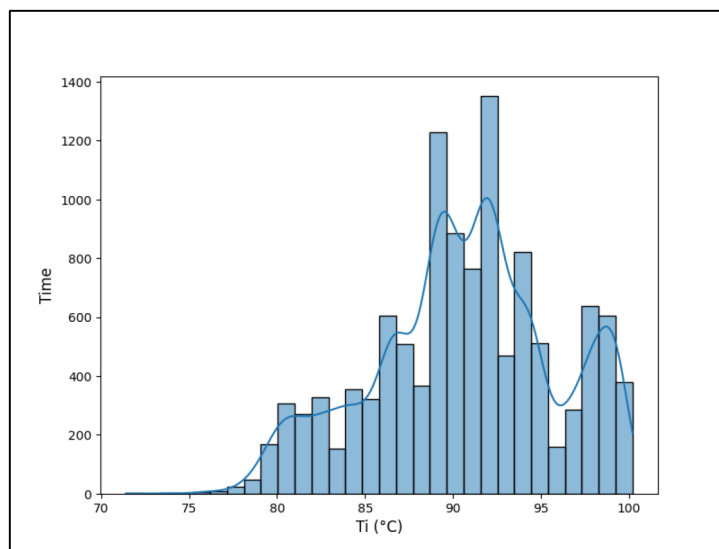


Figure 9—Histogram plot illustrating the distribution of intake temperature (T_i).

The results revealed that most features are non-normally distributed, deviating from the bell-shaped curve and exhibiting various patterns such as skewness, bimodality, or uniform distributions. These findings were later considered during the data cleaning and pre-processing step.

Data Cleaning and Pre-processing

The original dataset contained 24,000 data points, which were reduced to 12,000 after cleaning. The cleaning process varied depending on whether the data was used for prediction or classification tasks. Firstly, the steps in data preparation for the prediction models are as follows:

In general, it is better to deal with missing values first because there could be data loss to ensure data continuity. The dataset we are using contained lost downhole data when the sensor stopped recording. So forth, we had to remove any rows with more than five missing parameters, while keeping rows that have fewer missing values, and use a Forward Filling method (LOCF) to fill each missing datapoint with the last value known. This approach was identified as the most suitable imputation method for this case, considering the nature of the dataset, particularly when handling temporal data.

Duplicate data can skew analysis results and misrepresent the true nature of the dataset. Therefore, the duplicate rows were removed prior to doing data analysis.

The Interquartile Range (IQR) method was employed to detect potential outliers within the dataset. When outliers were identified, visualization methods—such as box plots shown in Figures 10 and 11 for samples of P_d and Q —were used to confirm the results and support decision-making regarding their treatment. According to domain knowledge, some extreme values hold significant information that is essential for building the model. Hence, they were capped using the Min-Max capping method (Winsorization) to retain them in the dataset while limiting their influence. However, erroneous outliers were removed completely (In some wells, CHP was recorded as a zero value). This comprehensive data cleaning approach ensured that the dataset was accurate, consistent, and suitable for the subsequent modeling stages, particularly for the implementation of the support vector regression model, as it is sensitive to outliers and null values.

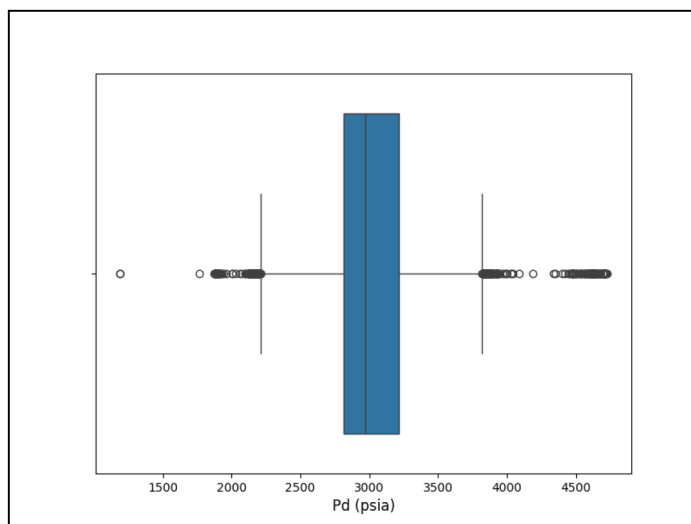


Figure 10—Sample box plot showing potential outliers in discharge pressure (P_d).

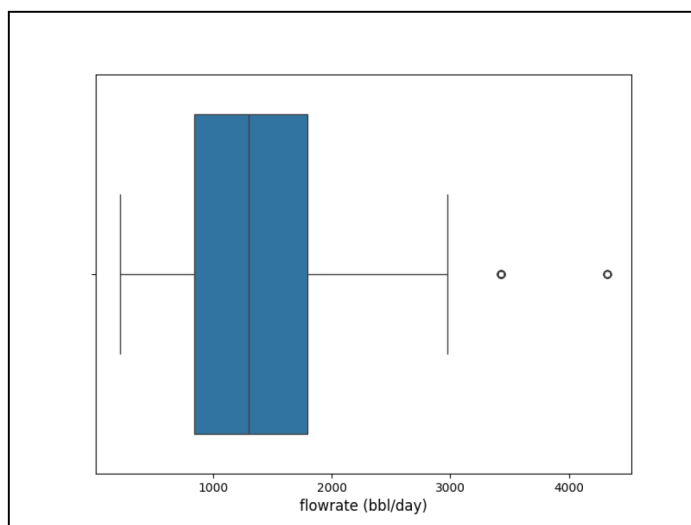


Figure 11—Sample Box Plot showing Potential Outliers in Flowrate (Q).

Moreover, the data were split into training and test datasets prior to feature scaling to prevent data leakage. This approach enhances the model's ability to generalize to unseen data by isolating the test data before any scaling. Consequently, the data were divided into 80% for training and 20% for testing. Since feature scaling is a critical preprocessing step for machine learning algorithms utilizing distance-based computations—such as Support Vector Regression (SVR)—applying scaling was vital. After evaluating available methods, standardization was selected as the most appropriate technique for this application. This method adjusts all features to have a mean of 0 and a standard deviation of 1, thereby normalizing their scale for consistent comparison prior to model training.

Secondly, the steps in data preparation for the classification model are as follows:

Primarily, null values and errors—such as zero values in certain features—were removed from the dataset. Unlike in the predictive models, for rows with fewer than five missing parameters, the backward filling method was preferred, where missing values are filled using the next available datapoint. Random Forest is known for its robustness in handling outliers, numerical features with varying scales, and high-dimensional datasets. Therefore, pre-processing—including feature scaling—was omitted, saving time and effort.

The primary objective of building the classification model is to predict failures of the ESP pump. To achieve this, a systematic approach was employed, involving the manual labeling of each failure event throughout the lifespan of the studied ESPs based on domain knowledge. Subsequently, one-hot encoding was applied to convert the categorical data into a format suitable for the model, with 0 for non-failure and 1 for failure. The detection focused on seven main ESP system failures: Overhead Transmission Line (OHTL) trip, Variable Speed Drive (VSD) or Diesel Generator (DG) trips, Flowline Pressure (FLP) fluctuation, reservoir issues (low productivity, high water cut, etc.), downhole equipment trip, and tubing blockage. At a later stage of modeling, feature engineering was applied, and new features were added to the dataset based on their importance, including pressure differential ($\Delta P = P_d - P_i$), thermal stress ($\Delta T = T_m - T_i$), electrical power ($power = ESP \text{ voltage} \times ESP \text{ Amps}$), and flow efficiency ($efficiency = \frac{Q}{power}$). Additionally, the run time and down time features were specifically added to the classification model's dataset, as they strongly contribute to predicting ESP failure.

Models Building, Optimization, and Evaluation

At this stage, the prepared data from previous steps was used to build and train various machine learning models, followed by performance evaluation using evaluation metrics. As mentioned, the models developed include Support Vector Regression (SVR) and Extreme Gradient Boosting Regression (XGBR) for prediction tasks, as well as Random Forest for classification tasks.

Initially, model development was conducted using Python Programming Language within integrated development environments (IDEs) such as Google Colaboratory and Visual Studio Code (VS Code), facilitating efficient implementation. The steps for each model are further discussed in the following section:

Generally, SVR was developed as two separate models: one for predicting discharge pressure and the other for predicting flowrate. As mentioned earlier, the dataset was initially split into 80% for training and 20% for testing as the common practice. After standardizing the dataset using the StandardScaler function from the scikit-learn library, the model's performance was optimized through extensive fine-tuning using the GridSearchCV technique. A total of 80 hyperparameter combinations were evaluated by varying the regularization parameter (C), kernel coefficient (gamma, γ), and epsilon (ϵ). Each combination was subjected to 5-fold cross-validation, resulting in 400 training iterations. This exhaustive search process identified the optimal hyperparameters: $C = 150$, $\gamma = 0.05$, and $\epsilon = 0.5$. Moreover, the Radial Basis Function (RBF) kernel was chosen for its ability to capture nonlinear relationships within the dataset. Lastly, the models were trained and tested separately for both P_d and Q , using standardized features to improve predictive accuracy.

In contrast, the XGBoost model is inherently robust to variations in feature scales and thus does not require feature scaling. The dataset was split using the same 80/20 ratio as in SVR—80% for training and 20% for testing—to ensure consistent evaluation. Unlike SVR, which required extensive hyperparameter tuning, XGBoost achieved optimal performance with only seven combinations of key parameters, significantly simplifying the tuning process. The selected configuration used a learning rate of 0.3, a maximum tree depth of 6, and a minimum child weight of 1, trained for 100 boosting iterations with a fixed random seed of 42 to ensure reproducibility. The models are then trained and tested for both P_d and Q .

Regarding failure identification in ESP systems, several models with different configurations and modeling techniques were developed. After evaluating their performance using metrics such as accuracy, precision, and recall, the optimal model that achieved acceptable results by correctly predicting nearly all failures was selected. As a standard practice, the dataset in all classification models was split into 80% for training and 20% for testing. The initial model utilized 100 decision trees; however, increasing the number to 200 in the second model did not enhance predictive performance. When the dataset exhibited severe class imbalance, the third model maintained the same number of trees (200) while incorporating oversampling techniques such as Synthetic Minority Oversampling Technique (SMOTE) and Adaptive

Synthetic Sampling (ADASYN) to address class imbalance, resulting in comparable yet modest accuracy improvements. Additionally, under-sampling of the non-failure data using the NearMiss technique was tested but it resulted in significantly lower accuracy. Lastly, the optimal model combined 100 trees with bootstrap aggregation (bagging) and an adjusted classification threshold of 0.75, achieving the most accurate failure prediction among the tested configurations.

Results and Discussion

This study introduces an innovative approach to optimize the performance of Electrical Submersible Pumps (ESPs) by developing multiple predictive models for key ESP system parameters and pump failure identification. The models include Support Vector Regression (SVR), Extreme Gradient Boosting (XGBoost), and Random Forest algorithms.

The performance of each model was evaluated using standardized metrics, with visualizing the results through graphs and charts to enhance interpretability. For prediction models, evaluation was based on three key metrics: the Root Mean Squared Error (RMSE), the coefficient of determination (R^2), and Average Absolute Percentage Error (AAPE). A comparative analysis was then conducted to identify the most effective model for predicting discharge pressure (P_d) and flowrate (Q). On the other hand, the classification model's performance was assessed using Precision, Recall, and F1-score, supplemented by a confusion matrix for detailed error analysis.

This structured evaluation ensures a comprehensive assessment of model accuracy and reliability in optimizing ESP performance. The following subsections provide detailed insights into each element, supported by field data, literature review, and visual interpretation.

Support Vector Regression (SVR) Model Performance and Evaluation

The Support Vector Regression (SVR) model was implemented to predict Discharge Pressure (P_d) and Flowrate (Q) using the pre-processed field data. Then it was evaluated for both parameters using Root Mean Squared Error (RMSE), coefficient of determination (R^2), and Average Absolute Percentage Error (AAPE).

For discharge Pressure (P_d), the SVR model yielded an RMSE of 60.33 and an R^2 of 0.972 on the training set. This indicates a high level of model accuracy, where 97% of the variation in P_d was explained by the model. On the testing set, the model showed strong performance with an RMSE of 76.28 and an R^2 of 0.955, signifying minimal decrease in its accuracy when applied to unseen data. The RMSE value indicates the average deviation of the predictions from the actual values, a smaller value signifies closer predictions to the true values. While R^2 describes the proportion of variation explained by the model, a value close to 1.0 suggests that the model is able to capture most of the variation and closely mirrors the actual values.

Additionally, the AAPE values provide a more interpretable measure of model performance, and enhance evaluation by presenting prediction error as a proportion of actual values. The model obtained AAPE values of 0.85% and 1.01% on training and testing sets, respectively, indicating that the average prediction deviates by less than 1.1% from the actual discharge pressure values. Moreover, the low AAPE values indicate precision and consistency in the model predictions for both seen and unseen data. These results are further validated and visualized in [Figure 12](#), where the actual and predicted P_d values are cross-plotted for the training and testing datasets. In this figure, the distribution of the points close to the 45° diagonal line signifies that the SVR model can predict the discharge pressures that closely approximate the actual discharge pressures, but with few observed outliers.

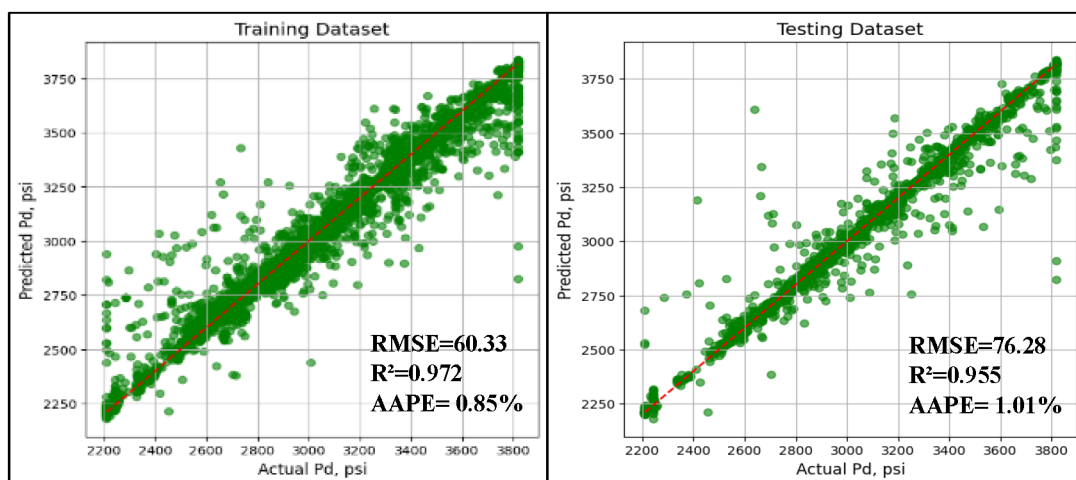


Figure 12—Crossplots of actual and predicted discharge pressure (P_d) using the SVR model for the training and testing datasets.

When applied to flowrate (Q), the SVR model achieved an RMSE of 151.53 and an R^2 of 0.925, indicating a solid fit on the training data. Testing performance showed only a slight decline, with an RMSE of 157.25 and an R^2 of 0.917. These results suggest the model effectively generalized to new data. Moreover, the model yielded an AAPE of 5.51% on the training set and 5.92% on the testing set. These percentages imply that, on average, the predicted flow rates deviated by less than 6% from the actual values. While higher than the corresponding AAPE values for P_d , the Q predictions still demonstrate acceptable accuracy, given the complex and dynamic nature of multiphase flow behavior in ESP-operated wells, reflecting the model's effective generalization capability.

The actual vs. predicted plots for Q in Figure 13 confirm the results, as the majority of predictions fall near the diagonal with limited dispersion.

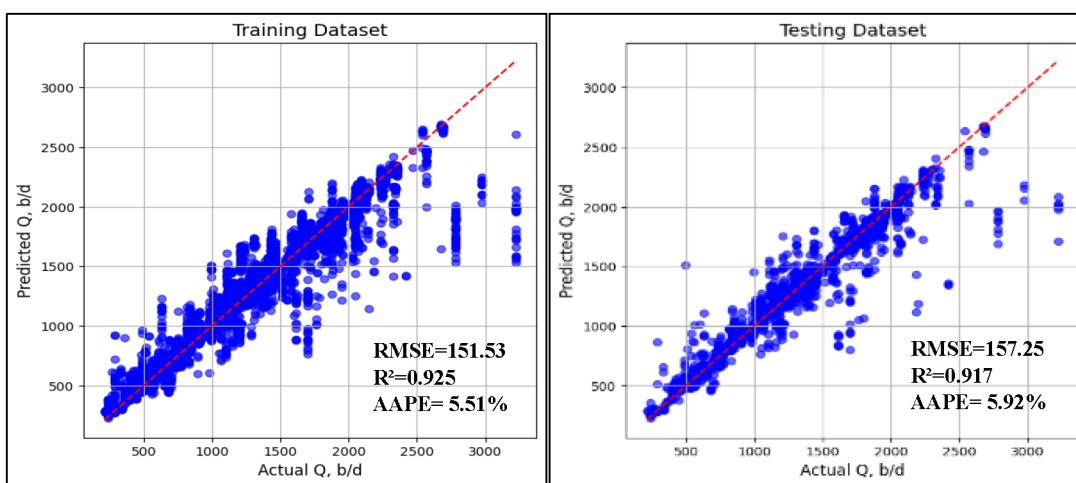


Figure 13—Crossplots of actual and predicted flowrate (Q) using the SVR model for the training and testing datasets.

Overall, the SVR model required extensive hyperparameter tuning and was sensitive to both feature scaling and outliers. Nevertheless, it demonstrated robustness in capturing non-linear relationships and maintained stable performance when predicting both P_d and Q .

Extreme Gradient Boosting (XGBoost) Model Performance and Evaluation

For P_d , the XGBoost model achieved a training RMSE of 19.63 and an R^2 of 0.977, indicating near-perfect prediction of the training data. Testing results remained robust with an RMSE of 64.16 and an R^2 of 0.968, reflecting excellent generalization capability. In practical terms, these metrics indicate that the model not only minimized the average prediction error but also captured nearly all the variability present in the dataset.

Furthermore, the model achieved AAPE values of 0.41% and 0.84% on the training and testing sets, respectively. These low values highlight the model's ability to predict P_d with high accuracy.

As visualized in Figure 14, the predicted values for P_d closely track the actual values for both training and testing datasets. The figure reveals the predicted values are closely packed along the ideal diagonal, which further validates the reliability of XGBoost in modeling pressure-related behaviors of ESP systems.

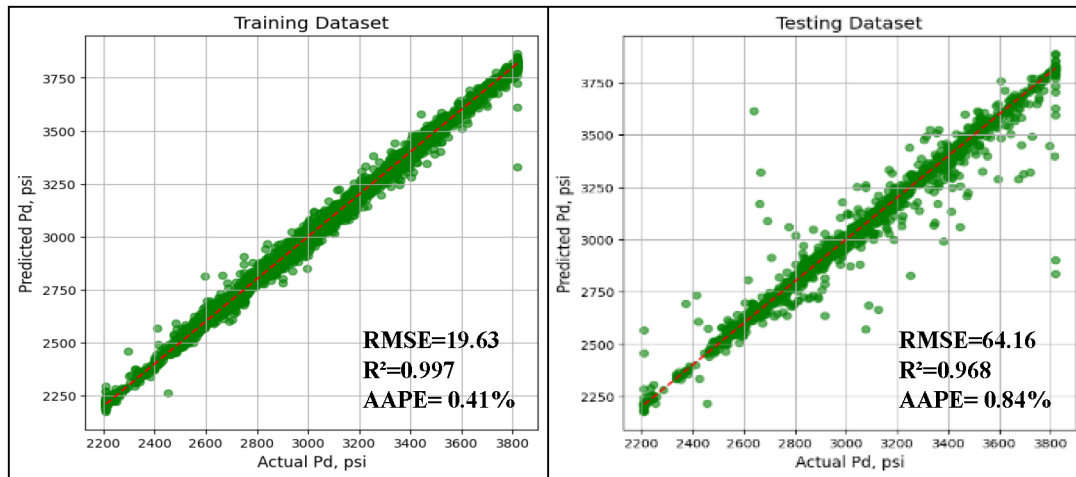


Figure 14—Crossplots of XGBoost model performance for discharge pressure prediction, showing R^2 metric for the training and testing datasets.

In predicting the flowrate (Q), the XGBoost model maintained its high level of accuracy, with an RMSE of 29.60 and an R^2 of 0.977 on the training set. On the testing set, the model achieved an RMSE of 98.16 and an R^2 of 0.968, outperforming SVR across all evaluation metrics.

The model also obtained AAPE values of 1.69% and 3.48% on the training and testing sets, respectively. These small percentages indicate the model's capability to generate predictions that remain close to the actual flow rate values, including for unseen data. From a practical perspective, they show that on average the deviation of the model with respect to actual values is small which endorses the ability of the model to capture complicated flow rate patterns in ESP systems among data sets. These values indicate not only a strong model fit but also an impressive capability to forecast flow rate behavior under varying operational conditions.

As shown in Figure 15, the actual versus predicted plots for Q further validate the model's performance, illustrating close agreement between the observed and predicted values.

Overall, XGBoost consistently delivered low error and high R^2 scores without extensive tuning, making it particularly well-suited for real-world applications. It efficiently manages large datasets, accommodates non-linear interactions, and handles imperfections in field data.

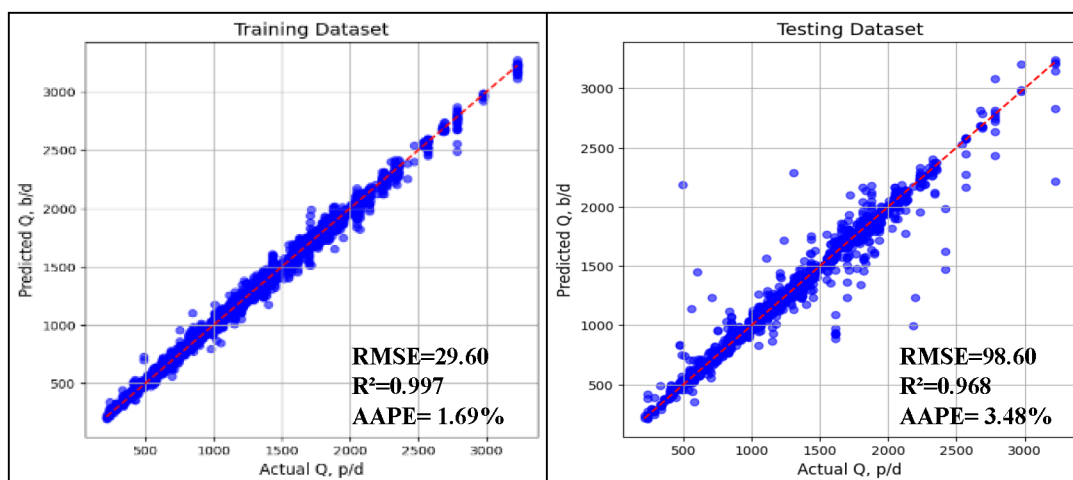


Figure 15—Crossplots of XGBoost model performance for flowrate prediction, showing R^2 metric for training (left) and testing (right) datasets.

Comparative Analysis of Prediction Models

A comprehensive comparative analysis was conducted between the Support Vector Regression (SVR) and Extreme Gradient Boosting Regression (XGBR) models to evaluate their predictive capabilities for discharge pressure (P_d) and flowrate (Q) within Electrical Submersible Pump (ESP) systems. The primary emphasis was placed on the models' testing performance, as these results provide a more realistic assessment of the models' ability to generalize to unseen operational data. The evaluation employed three key performance metrics: root mean squared error (RMSE), the coefficient of determination (R^2), and average absolute percentage error (AAPE).

To support this analysis, Figures 16 through 18 present visual comparisons of the models' performance in predicting both P_d and Q . These figures provide a straightforward interpretation of predictive accuracy and error distribution, with a primary focus on testing results. The R^2 performance of both models (SVR and XGBoost) is shown in Figure 16. During testing, XGBoost consistently achieved superior results, obtaining R^2 values of 0.968 for both P_d and Q , indicating strong predictive accuracy on unseen data. The SVR model, while still performing well, yielded slightly lower testing R^2 values of 0.955 for P_d and 0.917 for Q . Though the differences are modest, they demonstrate XGBoost's greater consistency in capturing variance in discharge pressure and flowrate predictions during testing.

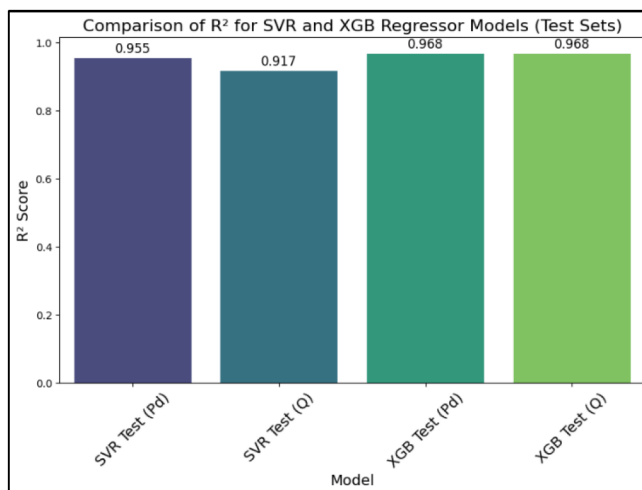


Figure 16—Bar chart comparing the R^2 values of SVR and XGBoost models for P_d and Q in the testing dataset.

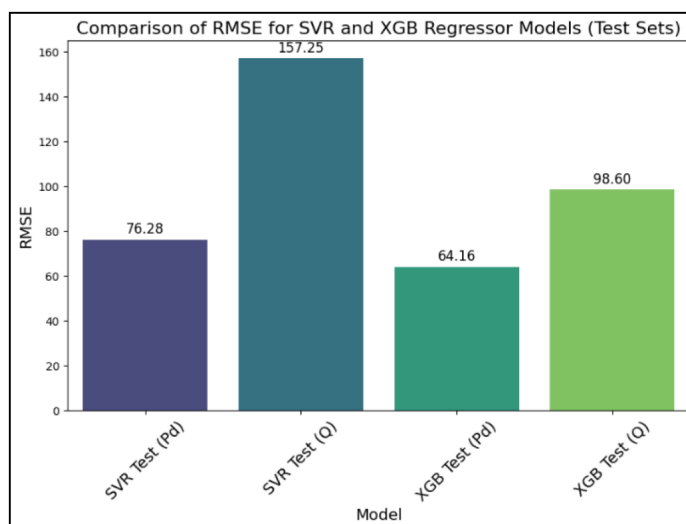


Figure 17—Bar chart comparing the RMSE values of SVR and XGBoost models for P_d and Q in the testing dataset.

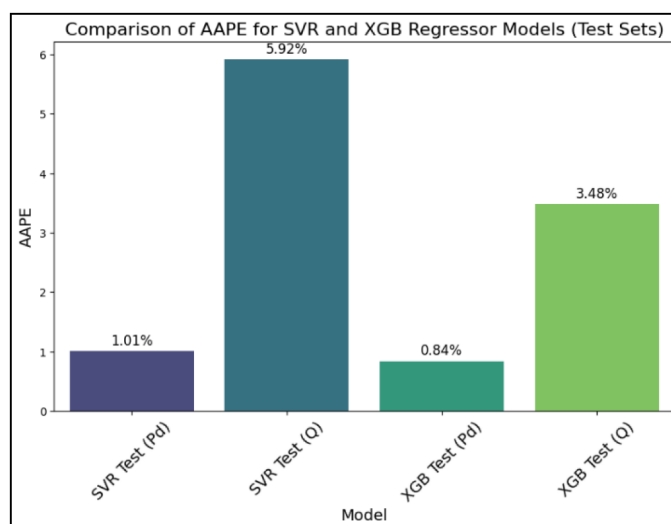


Figure 18—Bar chart comparing the AAPE values of SVR and XGBoost models for discharge pressure and flowrate in the testing dataset.

A clearer contrast between the models is observed in terms of RMSE values, as shown in Figure 17. On the testing data for P_d , XGBoost achieved a lower RMSE of 64.16, outperforming SVR, which recorded 76.28. This superiority was even more pronounced for Q , where XGBoost achieved an RMSE of 98.16 compared to SVR's significantly higher error of 157.25. The consistently higher RMSE values observed for Q across both models reflect the greater complexity inherent in flow rate prediction. Flowrate is influenced by dynamic multiphase flow phenomena, gas-liquid interactions, and variations in fluid properties such as viscosity, making it more challenging to model accurately. Discharge pressure, by contrast, exhibits greater stability and less variability, leading to lower prediction errors for both models.

A greater variance in model performance is also evident when considering the AAPE (average absolute percentage error) in Figure 18. Essentially, AAPE indicates the average magnitude of error as a percentage, making it valuable for comparing variables with different units. For P_d , XGBoost outperformed SVR with an AAPE of 0.84% compared to 1.01%, indicating more accurate predictions. The difference is even clearer when forecasting Q , where XGBoost achieved an AAPE of 3.48%, noticeably lower than the 5.92% recorded by SVR. These results show that XGBoost not only produced smaller absolute error but also delivered greater consistency and reliability in percent-based accuracy, particularly under the more

challenging conditions of flow rate forecasting. e lower AAPE values highlight XGBoost's superior ability to generalize across varying operational conditions, establishing it as the more resilient model for predicting ESP parameters.

In summary, the comparative evaluation clearly indicates that, although SVR maintains acceptable predictive performance, XGBR consistently outperforms the SVR, particularly under testing conditions where generalization to unseen data is essential. These findings are consistent with previously published studies (e.g., [Sathiparan et al., 2023](#)), which highlight the applicability of gradient boosting algorithms in complex engineering systems. The improved prediction accuracy observed for flowrate, in particular, holds significant operational value for ESP performance management, enabling better-informed maintenance planning, early fault detection, and more efficient resource allocation.

Pump Failure Classification Using Random Forest

To identify failures in ESP systems, several classification models with varying configurations were developed and assessed. These included adjustments in the number of decision trees and classification threshold, bagging usage, class imbalance handling techniques, and feature engineering. As presented in [Table 3](#), all models were grouped into two categories, under which they were tested and evaluated. In the first category, the models struggled with severe imbalance in data distribution, highlighting the need for resampling techniques such as the Synthetic Minority Oversampling Technique (SMOTE) for oversampling of the failure datapoints and the NearMiss technique for under-sampling of the non-failure datapoints. Although this approach showed superior performance in detecting the minority class (failure) compared to the original model, it still missed a significant portion of actual failure events.

Table 3—Classification model performance comparison

	Model	Sampling Method	Trees	Accuracy	Precision (Failure)	Recall (Failure)	F1-Score (Failure)
Before Data Balance	1	Original Model	100	0.99	0.67	0.11	0.18
	2	SMOTE	200	0.99	0.56	0.26	0.36
	3	Hybrid (SMOTE & NearMiss)	200	0.99	0.67	0.32	0.43
After Data Balance	4	Bagging	100	0.93	0.35	0.98	0.52
	5	Adjusted Threshold (0.75)	100	0.96	0.52	0.97	0.67

In contrast, the second category primarily utilized a balanced dataset to avoid such limitations. The models employed hyperparameter tuning with GridSearchCV, bagging, and adjusted thresholds to further improve results. Bagging involved creating multiple models on different subsets of the training data and combining their predictions. However, when used alone, the model flagged failures aggressively and successfully captured every true failure, but it overreacted, labeling many non-failures as failures and

resulting in a precision of just 0.35. This raised the need for threshold tuning to improve the precision-recall balance. Carterette (2009) noted that, in machine learning, precision and recall represent an inherent trade-off: increasing one often reduces the other. Precision focuses on limiting false positives, while recall seeks to minimize false negatives. Shifting the decision threshold can change this balance, but improving both simultaneously is generally not possible.

To provide deeper insight into the model's performance, errors, and weaknesses, a confusion matrix was used. The confusion matrixes in Figure 19 shows that the model correctly classified 2,823 non-failure cases and 6 failure cases. It misclassified 3 non-failure cases as failures (false positives) and failed to detect 13 actual failures (false negatives). These results indicate that the model performs very well in identifying non-failure events, with a very low false positive rate (low recall). However, while it captures some failure cases, a significant portion remains undetected. This reflects the challenge of classifying rare events in imbalanced datasets, despite applying both SMOTE and NearMiss techniques. In contrast, Figure 20 shows that the model delivered excellent failure prediction, correctly identifying 60 out of 62 actual failures, it also misclassified 56 non-failure cases as failures (false positives). The model achieved a high recall of 0.97, successfully detecting most failure events. Threshold tuning raised precision to 0.52 while maintaining an overall accuracy of 0.96, reflecting strong performance.

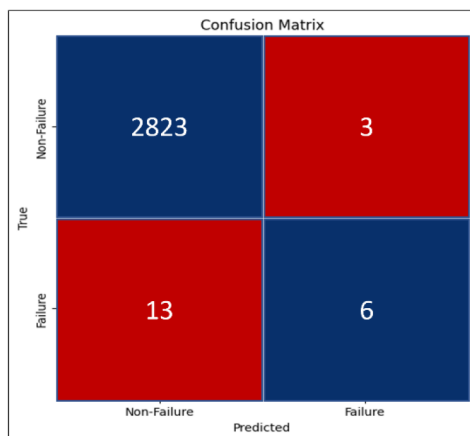


Figure 19—Confusion matrix of the hybrid classification model.

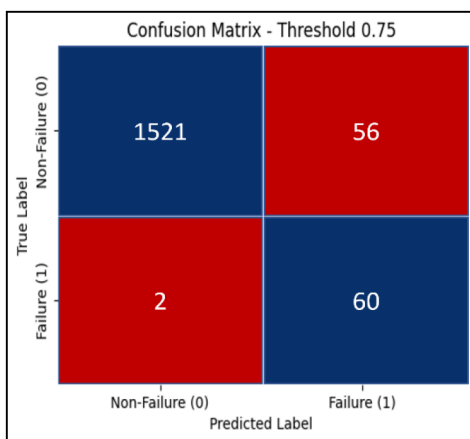


Figure 20—Confusion matrix of the adjusted threshold model.

Conclusions

This study investigated the performance of Electrical Submersible Pumps (ESPs) in an Arab oil field, integrating domain knowledge with data-driven approaches to address operational inefficiencies and

maintenance challenges, with a focus on the impact of reservoir characteristics, fluid properties, and operational parameters on pump efficiency and reliability. The findings confirmed the initial hypothesis that discharge pressure and flowrate are the most influential factors in ESP performance, thereby aligning with the study's primary objectives. The following points summarize the key conclusions:

- 1. Analysis of data from 44 wells across various carbonate formations revealed that discharge pressure (P_d) and flowrate (Q) are the most critical parameters affecting ESP performance, as they directly influence the pump's fluid-lifting capacity and ability to maintain optimal operating conditions.
- 2. Machine learning models (Support Vector Regression [SVR] and Extreme Gradient Boosting [XGBoost]) were successfully applied to predict discharge pressure and flowrate with XGBoost achieving better performance ($R^2 = 0.968$).
- 3. Those models were built on a strong foundation using 12,000 datapoints, including 18 critical ESP parameters.
- 4. The results of regression models support real-time monitoring of flowrate and enable data-driven decision-making without reliance on physical flowmeters.
- 5. Comparative evaluation of the predictive machine learning models revealed their respective strengths and reliability, validating their application in forecasting ESP behavior under different operational scenarios.
- 6. The classification model effectively predicted ESP failures with high accuracy of 0.96 and therefore is considered reliable for real-world applications.

Nomenclature

Parameter Notation	Parameter Definition	Unit
C	Regularization parameter	Same as target variable
P_d	Pump discharge pressure	psi
P_i	Pump intake pressure	psi
Q	Fluid flowrate	bbl/day
T_i	Intake temperature	°C
T_m	Motor temperature	°C
γ	Kernel coefficient	$1/(\text{input units})^2$ (often dimensionless after scaling)
ϵ	Epsilon-insensitive loss function	Same as target variable
ΔP	Pressure differential	psi
ΔT	Thermal stress	°C

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